FORWARD DEPLOYED ENGINEERING

Study Notes: Lecture 01 - Course Orientation, AI Workers & Architecture

1. Executive Summary

- 3-Month Strategic Roadmap: The course sequence focuses on achieving recognized international certifications (Associate & Architect), building reusable plugins, and publishing them on global marketplaces to sell internationally.
- AI Workers vs. Chatbots: Moving past read-only chatbot responses, AI workers execute end-to-end tasks inside dedicated cloud virtual machines (VMs) using skills and connectors.
- Core Governance Frameworks: AI operations are governed by authoritative Systems of Record, split into KSOR (Knowledge System of Record for policies and knowledge) and DSOR (Data System of Record for permissions and data governance).
- Modular Ecosystem: Reusable capabilities are packaged as Plugins combining MCP (Model Context Protocol) connectors for external access and Agent Skills for procedural consistency.

2. Core Concepts & Definitions

Term / Concept	Definition / Explanation
AI Worker	An autonomous agent that plans, executes commands, uses tools, operates in a VM, and creates editable digital artifacts.
Plugin	A packaged, reusable AI capability combining connectors, skills, and deterministic workflows.
Connector / MCP	Mechanism using Model Context Protocol (MCP) to safely connect AI workers to external tools, APIs, and data sources.

Agent Skill	Procedural instructions and rules that dictate *how* an AI worker performs a repeatable task consistently.
KSOR (Knowledge System of Record)	The authoritative source for organization knowledge, rules, and policies that the AI worker should follow.
DSOR (Data System of Record)	The governance layer controlling allowed connections, permissions, record-keeping, and action refusals.
Virtual Machine (VM) & Session	An isolated cloud computing environment allocated for a session where the AI worker runs commands and processes data.
Artifact	A durable, permanent output (documents, code, project files) that can be exported and saved beyond the session lifecycle.

3. Structured Body Hierarchy

- **Course Roadmap & Certifications**
    - 3-Step Progression: Credibility via certification Product building (plugins) International marketplace distribution.
        - Certification Tiers:
        - *FD 1001 (Associate Foundations)*: Focuses on practical problem-solving using AI tools ($125 exam fee). Requires passing a tougher internal exam first.
        - *FD 102 (Architect Foundations)*: Advanced fast-track track focusing on system architecture, agents, and plugins.
- **AI Workers, VMs, & Cloud Sessions**

  - Autonomous Execution: Unlike chatbots that merely explain steps, AI workers plan and execute tasks autonomously.
  - Cloud Persistence: Work continues on cloud VMs even if the user closes their laptop; results are retrieved later.
  - Artifacts vs. Temporary Files: Session-dependent temporary files disappear when the VM expires, whereas artifacts are durable, exportable deliverables.

- **Plugins & Marketplace Strategy**
  - The "Piggy Bank" Strategy: Publishing plugins on established marketplaces leverages existing platform audiences rather than self-marketing from scratch.
  - Architecture: Connector (Access) + Skill (Procedure) = Reusable Plugin.
- **Governance & Standards: KSOR, DSOR, & MCP**
  - KSOR: Ensures AI workers rely on validated organizational knowledge and policies rather than unverified web data.
  - DSOR: Enforces strict permission boundaries (e.g., allowing read/invoice creation while blocking unauthorized data deletion).
  - MCP Standards: Facilitates standardized interoperability between AI systems and external business systems.
- **Interface Controls & Permissions**
  - Approval Modes: Manual approval (user reviews and approves actions) is recommended for beginners before moving to auto-approval.
  - UI Controls: Features model selection, effort/hardness configuration, routing controls, and the plus menu for tools and web search.

4. Review & Active Recall (Q&A)

- Q: What is the main difference between an AI chatbot and an AI worker?

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- A: A chatbot tells the user how to do something (read-only guidance), whereas an AI worker executes the work, runs commands in a VM, uses tools, and generates deliverables.

- Q: How do KSOR and DSOR differ in function?

- A: KSOR defines *what* the AI should know and follow (knowledge/policies), while DSOR governs *what* the AI is permitted to do with real systems (permissions/connections).

- Q: Why are artifacts distinguished from temporary files?

- A: Temporary files exist only during the active cloud session and disappear when the VM is destroyed, whereas artifacts are durable, permanent outputs that can be saved and exported.

- Q: What is the strategic benefit of the "Piggy Bank" marketplace approach?

- A: It allows developers to list their built plugins on established platforms, instantly gaining access to an existing international audience and user base without heavy independent marketing.